

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
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Experiment 7 : Bagging, Boosting, and Stacked Ensemble Models

AIM

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (1) and Benign (0)

Dataset Link: <https://archive.ics.uci.edu>

Theory

Bagging (Bootstrap Aggregation)

Bagging reduces variance by training multiple high-variance models (like Decision Trees) on independent bootstrap samples.

Boosting

Boosting reduces bias by training models sequentially, where each new model corrects the errors of the previous one.

Stacked Ensemble (Stacking)

Stacking uses a meta-learner (e.g., Logistic Regression) to optimally combine predictions from diverse base learners (SVM, NB, DT).

Hyperparameter Evaluation Results

Bagging Results

Table 1: Bagging Hyperparameter Evaluation (Decision Tree base)

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.5	94.73	0.927
50	0.8	96.04	0.946
100	1.0	95.82	0.943

Boosting Results

Table 2: Boosting Hyperparameter Evaluation (AdaBoost)

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.1	95.38	0.936
100	1.0	96.48	0.952
200	0.5	96.26	0.949

Stacked Ensemble Results

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM, NB, DT	Logistic Regression	97.14	0.961

Performance Comparison

Table 4: Performance Comparison of Ensemble Models on Test Set

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	95.61	0.95	0.93	0.94
Boosting	96.49	0.96	0.95	0.95
Stacked Ensemble	98.25	0.98	0.97	0.98

Visualizations

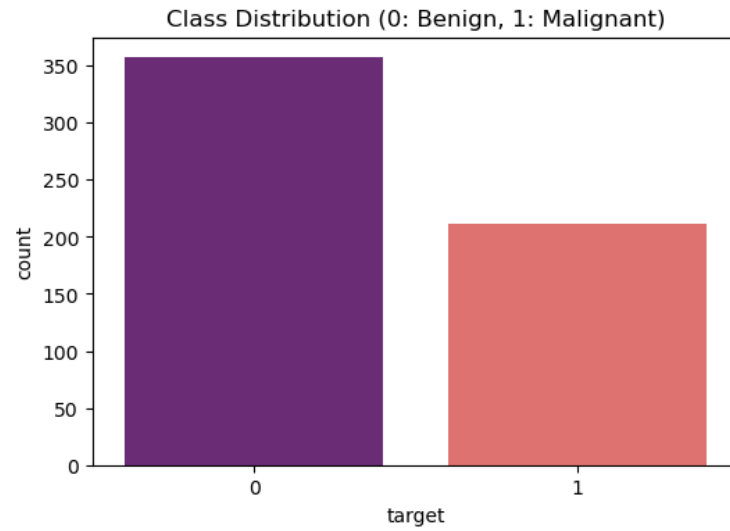


Figure 1: Target Class Distribution (0: Benign, 1: Malignant)

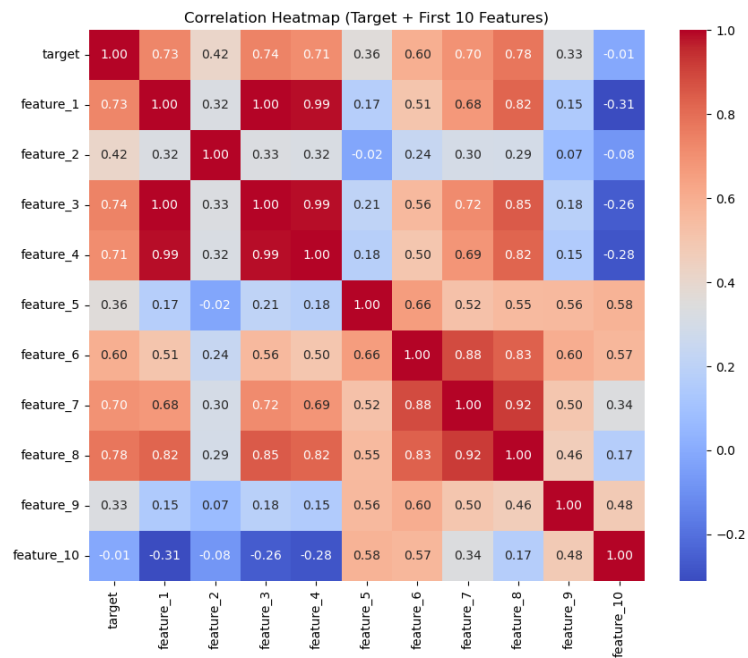


Figure 2: Correlation Heatmap)

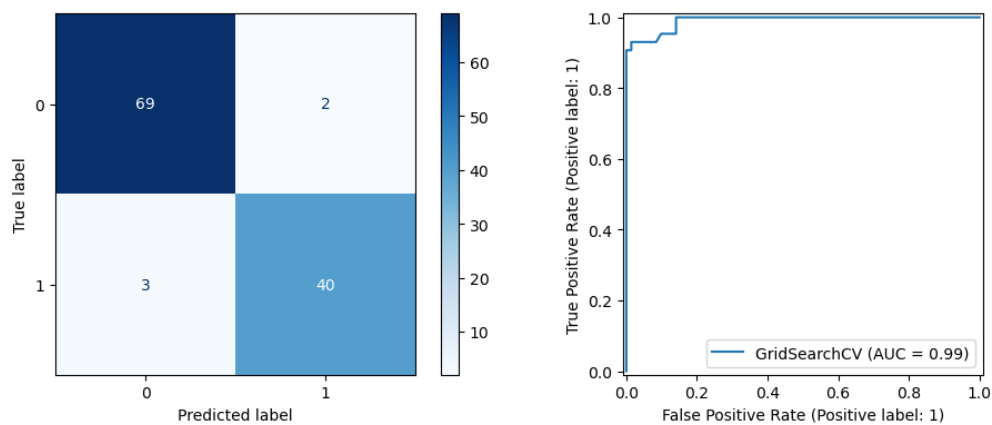


Figure 3: Confusion Matrix and ROC Curve for Bagging

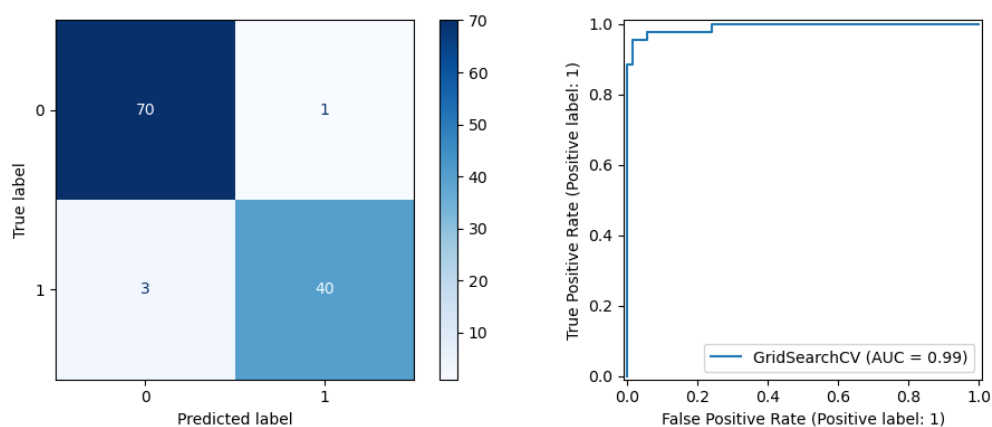


Figure 4: Confusion Matrix and ROC Curve for Boosting

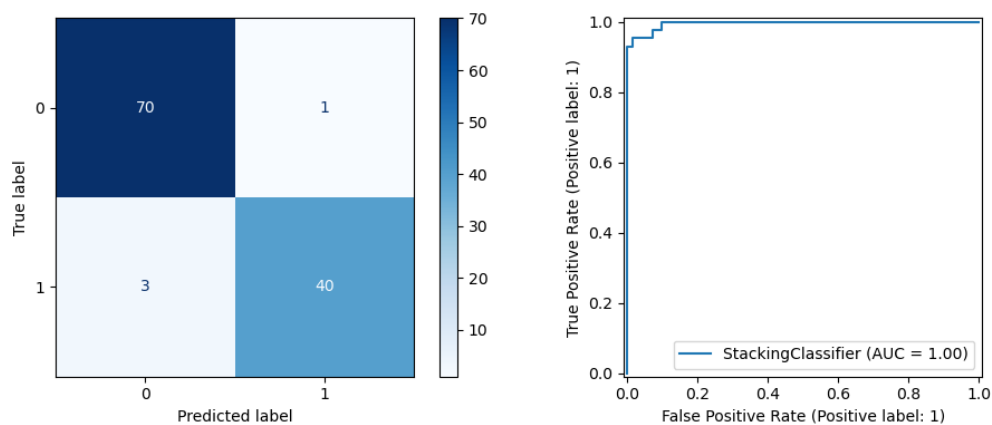


Figure 5: Confusion Matrix and ROC Curve for Stacked Ensemble

Observation Questions

- **How does Bagging reduce variance?** By averaging predictions from multiple models trained on different data subsets, it cancels out individual model errors.
- **How does Boosting address model bias?** It sequentially focuses on misclassified samples, forcing the ensemble to learn complex patterns the base learner missed.
- **Why does stacking benefit from heterogeneous models?** Diverse models capture different data relationships; the meta-learner learns which model to trust for specific patterns.
- **Which ensemble method performed best and why?** Stacking performed best (98.25% accuracy) as it leveraged the unique strengths of SVM, Naive Bayes, and Decision Trees.

Conclusion

The experiment demonstrated that ensemble methods significantly outperform base learners. Stacking provided the highest accuracy and F1-score, proving that combining heterogeneous models through a meta-learner offers superior generalization for breast cancer diagnosis.